Ps 03

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<2026-04-01 Wed>

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1 Problem 30

1.1 Task

Prove that
(a) $\binom{2n}{n} = \frac{2^{2n}}{\sqrt{\pi n}}\left(1 + O\left(\frac{1}{n}\right)\right)$,
(b) $\binom{n}{k} \geq \frac{n^k}{4k!}$ if $k \leq \sqrt{n}$.

1.2 Solution

1.2.1 (a)

Stirling approximation:

$$n! = \sqrt{2\pi n}\left(\frac{n}{e}\right)^n \left(1 + O\left(\frac{1}{n}\right)\right)$$

$$\begin{aligned}
\binom{2n}{n} &= \frac{(2n)!}{(n!)^2} \\
&= \frac{\sqrt{4\pi n}\left(\frac{2n}{e}\right)^{2n}\left(1 + O\left(\frac{1}{n}\right)\right)}{\left(\sqrt{2\pi n}\left(\frac{n}{e}\right)^n\left(1 + O\left(\frac{1}{n}\right)\right)\right)^2} \\
&= \frac{2\sqrt{\pi n} \cdot 2^{2n}\left(\frac{n}{e}\right)^{2n}\left(1 + O\left(\frac{1}{n}\right)\right)}{2\pi n\left(\frac{n}{e}\right)^{2n}\left(1 + O\left(\frac{1}{n}\right)\right)^2} \\
&= \frac{2^{2n}}{\sqrt{\pi n}} \cdot \frac{1 + O\left(\frac{1}{n}\right)}{\left(1 + O\left(\frac{1}{n}\right)\right)^2} \\
&= \frac{2^{2n}}{\sqrt{\pi n}}\left(1 + O\left(\frac{1}{n}\right)\right)
\end{aligned}$$

1.2.2 (b)

$$\begin{aligned}
\binom{n}{k} &= \frac{n(n-1)\cdots(n-k+1)}{k!} \\
&= \frac{n^k}{k!} \cdot \prod_{i=0}^{k-1}\left(1 - \frac{i}{n}\right)
\end{aligned}$$

Using $\prod(1 - a_i) \geq 1 - \sum a_i$ for $a_i \in [0, 1]$:

$$\prod_{i=0}^{k-1}\left(1-\frac{i}{n}\right) \geq 1-\sum_{i=0}^{k-1}\frac{i}{n} = 1-\frac{k(k-1)}{2n} > 1-\frac{k^2}{2n} \geq 1-\frac{1}{2} = \frac{1}{2} \geq \frac{1}{4}$$

Hence $\binom{n}{k} \geq \frac{n^k}{k!} \cdot \frac{1}{4} = \frac{n^k}{4\,k!}$. $\square$

2 Problem 31

2.1 Task

You shuffle a deck of 10 cards, each marked with a distinct number from 1 to 10. You then remove 3 cards from the deck, one by one. What is the probability that the cards arrive in increasingly sorted order?

2.2 Solution

$$S = \{SML, SLM, MSL, MLS, LSM, LMS\}$$

With only one increasing event SML. Hence, $\Pr(SML) = \frac{1}{6}$.

3 Problem 32

3.1 Task

Let A, B, C be events, i.e., subsets of Ω. We call A and B conditionally independent given C, if

$\Pr(A \cap B \mid C) = \Pr(A \mid C)\Pr(B \mid C)$.

Give an example of three events A, B, and C so that A and B are not independent but are conditionally independent given C.

3.2 Solution

Let $\Omega = \{(F, HH), (F, HT), (F, TH), (F, TT), (B, HH)\}$, with $\Pr(F, \cdot) = \frac{1}{8}$ and $\Pr(B, HH) = \frac{1}{2}$.

Define $A =$ first flips heads, $B =$ second flips heads, $C =$ fair coin chosen.

$$
\begin{aligned}
A &= \{(F, HH), (F, HT), (B, HH)\} \\
B &= \{(F, HH), (F, TH), (B, HH)\} \\
C &= \{(F, HH), (F, HT), (F, TH), (F, TT)\}
\end{aligned}
$$

- Non independence of A, B:

$$
\begin{aligned}
A \cap B &= \{(F, HH), (B, HH)\}, \qquad \Pr(A \cap B) = \frac{1}{8} + \frac{1}{2} = \frac{5}{8} \\
\Pr(A) &= \underbrace{\frac{1}{8}}_{(F,HH)} + \underbrace{\frac{1}{8}}_{(F,HT)} + \underbrace{\frac{1}{2}}_{(B,HH)} = \frac{3}{4} \\
\Pr(B) &= \underbrace{\frac{1}{8}}_{(F,HH)} + \underbrace{\frac{1}{8}}_{(F,TH)} + \underbrace{\frac{1}{2}}_{(B,HH)} = \frac{3}{4} \\
\Pr(A)\Pr(B) &= \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{16} \neq \frac{5}{8} = \Pr(A \cap B)
\end{aligned}
$$

- Conditional independence given C:

$$
\begin{aligned}
\Pr(A \mid C) &= \frac{\Pr(A \cap C)}{\Pr(C)} = \frac{1/4}{1/2} = \frac{1}{2} \\
\Pr(B \mid C) &= \frac{\Pr(B \cap C)}{\Pr(C)} = \frac{1}{2} \\
\Pr(A \cap B \mid C) &= \frac{\{(F, HH)\}}{\Pr(C)} = \frac{1/8}{1/2} = 1/4 = \Pr(A \mid C)\Pr(B \mid C)
\end{aligned}
$$

4 Problem 33

4.1 Task

You are a fantasy role playing fan, and you know that the damage you cause with your fireball attack is $40d6$, which means that you have to roll 40 ordinary 6 sided dice and add up all the numbers (of course you could also roll the same 6 sided dice 40 times). Since this is rather tedious, you say that you would like to replace the rolling of the dice by the expected value. What is the expected value, and what is the probability that you are off by more than 1 when you actually do roll the dice?

4.2 Solution

Let $S = \sum_{i=1}^{40} X_i$ where X_i are i.i.d (independent and identical distributed) rolls of a fair die.

$$E[X_i] = \sum_{x=1}^{6} x \cdot \frac{1}{6} = \frac{7}{2}$$
$$E[S] = 40 \cdot \frac{7}{2} = 140$$

$$E[X_i^2] = \sum_{x=1}^{6} x^2 \cdot \frac{1}{6} = \frac{1+4+9+16+25+36}{6} = \frac{91}{6}$$
$$\mathrm{Var}(X_i) = E[X_i^2] - (E[X_i])^2 = \frac{91}{6} - \frac{49}{4} = \frac{35}{12}$$

$$\mathrm{Var}(S) = 40 \cdot \frac{35}{12} = \frac{350}{3}, \qquad \sigma_S = \sqrt{\frac{350}{3}} \approx 10.80$$

By CLT, $S \dot{\sim} N(140, \frac{350}{3})$. Standardizing:

$$\begin{aligned} P(|S-140|>1) &= 1 - P\left(|Z| \leq \frac{1}{\sigma_S}\right) \\ &= 1 - P(|Z| \leq 0.0926) \\ &= 1 - (2\Phi(0.0926) - 1) \\ &= 1 - (2 \cdot 0.5369 - 1) \\ &= 1 - 0.0738 \\ &\approx 0.926 \end{aligned}$$

The expected value is 140. The probability of being off by more than 1 is $\approx 92.6\%$.

```
N = 20_000_000
off = count(abs(sum(rand(1:6) for _ in 1:40) - 140) > 1 for _ in 1:N)
println("P(|S-140|>1)  $(round(off/N, digits=4))  (N=$N)")
```

5 Problem 34

5.1 Task

> Let X be a nonnegative random variable, and suppose that $E[X]$ is defined. Prove Markov's inequality
>
> $\Pr(X \geq t) \leq E[X]/t$
>
> for all $t \geq 0$.

5.2 Solution

Let $t > 0$ (the case $t = 0$ is vacuous since $E[X]/t$ is undefined).

Define $Y = t \cdot 1_{X \geq t}$. We claim $Y \leq X$ pointwise:

- If $X(\omega) \geq t$: $Y(\omega) = t \leq X(\omega)$.
- If $X(\omega) < t$: $Y(\omega) = 0 \leq X(\omega)$ since $X \geq 0$.

By monotonicity of expectation, $E[Y] \leq E[X]$. Since Y takes value t with probability $\Pr(X \geq t)$ and 0 otherwise, $E[Y] = t \cdot \Pr(X \geq t)$. Therefore:

$$\begin{aligned} t \cdot \Pr(X \geq t) &\leq E(X) \\ \Pr(X \geq t) &\leq \frac{E(X)}{t} \end{aligned}$$

6 Problem 35

6.1 Task

> Prove that the variance of the geometric distribution is q/p^2.

6.2 Solution

Define $p =$ probability of success, $\quad q = 1 - p$.

$$
\begin{aligned}
E[X] &= \sum_{k=1}^{\infty} kq^{k-1}p \\
&= \frac{p}{q}\sum_{k=0}^{\infty} kq^{k} \\
&= \frac{p}{q}\cdot\frac{q}{(1-q)^2} \\
&= \frac{p}{q}\cdot\frac{q}{p^2} \\
&= \frac{1}{p}
\end{aligned}
$$

$$
\begin{aligned}
E[X^2] &= \sum_{k=1}^{\infty} k^2q^{k-1}p \\
&= \frac{p}{q}\cdot\sum_{k=1}^{\infty} k^2q^{k} \\
&= \frac{p}{q}\cdot\sum_{k=1}^{\infty} (k(k-1)+k)q^{k} \\
&= \frac{p}{q}\cdot\left(\sum_{k=1}^{\infty} k(k-1)q^{k} + \sum_{k=1}^{\infty} kq^{k}\right) \\
&= \frac{p}{q}\cdot\left(\frac{2q^2}{(1-q)^3} + \frac{q}{(1-q)^2}\right) \\
&= \frac{p}{q}\cdot\left(\frac{2q^2+qp}{p^3}\right) \\
&= \frac{2q+p}{p^2} \\
&= \frac{2-p}{p^2}
\end{aligned}
$$

$$
\begin{aligned}
\mathrm{Var}[X] &= E[X^2] - (E[X])^2 \\
&= \frac{2-p}{p^2} - \frac{1}{p^2} \\
&= \frac{1-p}{p^2} \\
&= \frac{q}{p^2}
\end{aligned}
$$